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Dublin City University

AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1042 - Computer Networks 2

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,3

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer all questions.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 30]***

Q 1(a) **[20 Marks]**

The IP address 192.168.100.0 is to be subnetted. Using 2 bits for subnetting, answer the following questions:

1. What class is this address? Explain how you find it.
2. What is the default subnet mask of this class?
3. What is the new subnet mask of this address?
4. How many subnets can be created?
5. What are the IP addresses of these subnets?
6. What is the broadcast address of each subnet?
7. What are the first and last host IP addresses of each subnet?

Show all your work clearly and use full 4 bytes decimal-dot-notation throughout.

Q 1(b) **[10 Marks]**

Given the network address 145.10.0.0/14, answer the following:

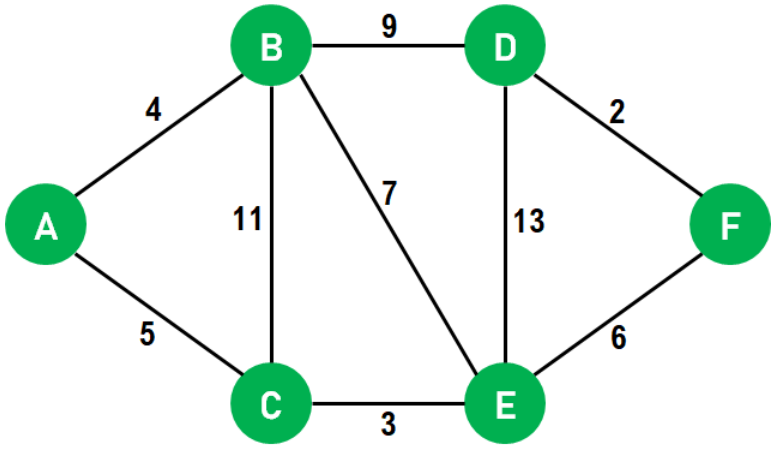
1. How many Class B networks are included in this supernet?
2. What are their IP addresses?
3. What is the broadcast address of this supernet?
4. What is the first and last valid host IP addresses in this supernet?

[End of Question 1]

QUESTION 2 ***[TOTAL MARKS: 34]***

Q 2(a) **[10 Marks]**

Using the graph below and following the Dijkstra's algorithm, show the order through which the nodes will be visited to compute shortest path from node A to node F.



Q 2(b) **[10 Marks]**

Explain the following security measures used to mitigate network attacks:

1. Defence-in-depth
2. Principle of least privilege
3. Separation of duties

Q 2(c) **[4 Marks]**

Angela, an IT staff member at ACME Inc., notices that communication with the company's web server is very slow. After investigating, she determines that the cause of the slow response is a computer on the internet sending a very large number of malformed web requests to ACME'S web server. What type of attack is described in this scenario?

Q 2(d) **[10 Marks]**

Write the lines to be added to the SAO record to create the following:

1. Type A record that maps the domain name "books.com" to the IP address 205.17.10.1
2. Type CNAME record for "drama" that points to "books.com"
3. Type NS record for "ns1", the nameserver responsible for resolving DNS queries of "books.com"
4. Type PTR record to enable reverse DNS lookup for "books.com"

[End of Question 2]

QUESTION 3 ***[TOTAL MARKS: 36]***

Q 3(a) **[10 Marks]**

The following firewall rules are to be implemented:

1. Allow established TCP connections to all IP addresses
2. Forbid all connections except those on this list
3. Allow access to the DHCP server only from within the network
4. Allow access to Apache web server on port 80

In what order would you place these rules to achieve an efficient drop policy.

Q 3(b) **[10 Marks]**

Explain the count-to-infinity problem encountered in Distance Vector Routing using a diagram.

Q 3(c) **[4 Marks]**

A company wants a cloud service solution that can provide a deployment platform for its applications without worrying about the infrastructure requirements. Which cloud service should the company go for?

Q 3(d) **[6 Marks]**

Abbreviate the following IPv6 address:

00ab:7500:0000:f2b0:bacf:0000:0000:fa5d

Q 3(e) **[3 Marks]**

Which of the following statements is true about containerization?

a. It enables users to run multiple operating systems on the hardware of a single device
b. It enables users to use applications online without the need to download them
c. It provides users with a development platform to implement their applications
d. It enables users to deploy multiple applications using the same operating system on a single VM or server

Q 3(f) **[3 Marks]**

Which routing protocol is used to connect ASs on the Internet?

a. RIPv2
b. EIGRP
c. OSPF
d. BGP

[End of Question 3]

[END OF EXAM]